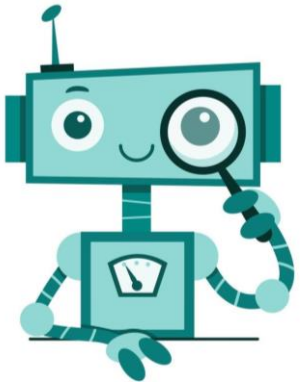
Artificial Intelligence (AI)



AGENDA

- Theory of Recurrent neural network
- Implementation of CNN (Practical)

Recurrent Neural Network

- A type of neural network commonly used in natural language processing, speech recognition, and time-series analysis.
- Unlike feedforward neural networks, which process input data in a single pass, RNNs process sequences of data by maintaining a "hidden state" that summarizes the information from previous time steps. This allows RNNs to capture the temporal dependencies and patterns in the input sequence.
- During training, the weights of the network are adjusted using backpropagation through time, which is a variant of backpropagation that considers the dependencies between time steps. The process of adjusting the weights is known as optimization.

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Types of RNN

- **LSTM:** LSTM stands for Long Short-Term Memory. It is a type of recurrent neural network (RNN) that is designed to handle the vanishing gradient problem in traditional RNNs, which occurs when gradients in backpropagation become too small to effectively train the network.
- **GRU:** GRU stands for Gated Recurrent Unit. It is a type of recurrent neural network (RNN) that is like the LSTM but has a simpler architecture.

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